

Sales Performance Analysis

Prepared by: Gabriel Andrew

Tools Used:

- Microsoft Excel
- Python (Pandas & Matplotlib)
- SQL Server
- Microsoft Power BI

1. Executive Summary

This project analyzed sales data from the Sample Superstore dataset to evaluate business performance. The analysis focused on identifying sales trends, product performance, regional sales distribution, and profitability. Insights from the analysis were presented using SQL queries, Python visualizations, and an interactive Power BI dashboard.

2. Business Problem

The management team wanted answers to the following questions:

- Which products generate the highest revenue?
- Which categories perform best?
- Which regions contribute the most sales?
- How do sales change over time?
- Which areas deserve more investment?

3. Dataset Description

The dataset contains retail sales transactions with information such as:

- Order ID
- Order Date
- Customer
- Region
- Category
- Product Name
- Sales
- Quantity
- Discount
- Profit

4. Data Cleaning

The dataset was cleaned using Python and Pandas.

The following steps were performed:

- Checked for missing values.
- Checked for duplicate records.
- Verified data types.
- Converted date columns to datetime format where necessary.
- Prepared the data for visualization and analysis.

5. Analysis

The following analyses were completed:

SQL

- Total Sales
- Total Profit
- Top 10 Products
- Top 5 Customers
- Sales by Category
- Sales by Region

- Monthly Sales
- Average Profit by Category

Python (Matplotlib)

Created:

- Monthly Sales Line Chart
- Sales by Category Bar Chart
- Top 10 Products Horizontal Bar Chart
- Sales by Region Pie Chart
- Profit by Category Bar Chart

Power BI

Created an interactive dashboard containing:

- Total Sales KPI
- Total Profit KPI
- Number of Orders KPI
- Monthly Sales
- Sales by Category
- Sales by Region
- Region Filter
- Category

6. Key Insights

- Technology generated the highest profit
- The West region recorded the highest sales
- Office Supplies had consistent sales
- Monthly sales varied throughout the year
- A small number of products contributed significantly to total revenue

7. Recommendations

- Increase investment in tech products
- Expand marketing efforts in the West region
- Review products with low profitability
- Monitor monthly sales trends to improve inventory planning

8. Conclusion

This project demonstrated how SQL, Python and Power BI can be combined to transform raw sales data into meaningful business insights. The analysis identified trends in sales, profitability and regional performance, enabling data-driven recommendations for improving business performance.